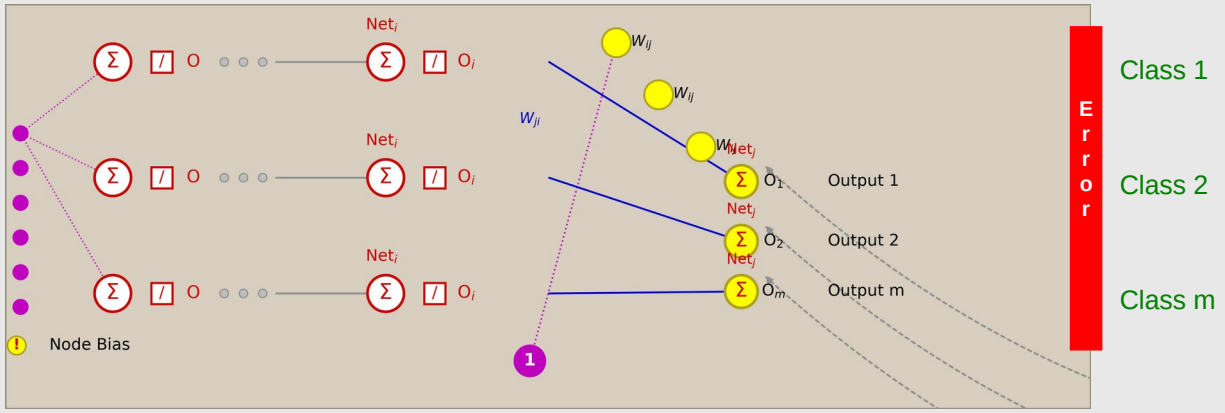
Gradient Descent

$$W_{ji}^{updated} = W_{ji}^{old} - \eta \frac{\partial E}{\partial W_{ji}}$$

$$W_{ji}^{updated} = W_{ji}^{old} + \Delta_{ji}$$

Where $\Delta_{ji} = -\eta \frac{\partial E}{\partial W_{ji}}$



$$E = \frac{1}{2}\sum_j (t_j - O_j)^2$$

For Sigmoid Activation

$$O_j = f(Net_j) = \frac{1}{1 + e^{-Net_j}}$$

$$Net_j = \sum_i (w_{ji} O_i)$$

$$\frac{\partial E}{\partial O_j} = -(t_j - O_j)$$

$$\frac{\partial O_j}{\partial Net_j} = O_j(1 - O_j)$$

$$\frac{\partial Net_j}{\partial W_{ji}} = O_i$$

$$\frac{\partial E}{\partial W_{ji}} = \frac{\partial E}{\partial O_j} \cdot \frac{\partial O_j}{\partial Net_j} \cdot \frac{\partial Net_j}{\partial W_{ji}} = \frac{\partial E}{\partial Net_j} \cdot \frac{\partial Net_j}{\partial W_{ji}}$$

$$= -(t_j - O_j) \cdot O_j(1 - O_j) \cdot O_i = -\delta_j \cdot O_i$$

Where $\delta_j = -\frac{\partial E}{\partial Net_j} = (t_j - O_j) \cdot O_j(1 - O_j)$ (δ_j depends on j ONLY)

$$\frac{\partial E}{\partial O_j} = -(t_j - O_j)$$

$$\frac{\partial O_j}{\partial Net_j} = O_j(1 - O_j)$$

$$\frac{\partial Net_j}{\partial W_{ji}} = O_i$$

$$\Delta_{ji} = -\eta \frac{\partial E}{\partial W_{ji}} = \eta \delta_j \cdot O_i \quad \textbf{Where} \quad \delta_j = (t_j - O_j) \cdot O_j(1 - O_j) \text{ for output Layer}$$